



Microbiology and Infection Control

FOR NURSING STUDENTS – LEVEL 1

Course Code: **MED 22121**

Prepared by:
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Dr. Rania Hamed

What is this course about?

This course introduces fundamental microbiological concepts related to nursing practice, including:

- ✓ Types of microorganisms
- ✓ Normal flora
- ✓ Routes of transmission
- ✓ Antimicrobial agents
- ✓ Hospital-acquired infections (HAIs)
- ✓ Sterilization and disinfection
- ✓ Infection control policies in healthcare settings.



Goal: Protect patients and healthcare workers from infections.



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Why Do Nurses Study Microbiology?

Nurses need microbiology to:

- ✓ Understand causes of infection
- ✓ Prevent infection spread
- ✓ Apply infection control measures
- ✓ Protect patients and themselves
- ✓ Interpret lab results
- ✓ Educate patients properly



Infection control is a daily nursing responsibility.



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Course Learning Outcomes (CLOs)

By the end of this course, students will be able to:

- ✓ **Classify microorganisms and describe their structure**
- ✓ **Apply sterilization and disinfection methods**
- ✓ **Explain the role of normal flora**
- ✓ **Interpret lab findings of infections**
- ✓ **Apply infection control guidelines**
- ✓ **Use reliable infection control resources**



Infection control is a daily nursing responsibility.



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Main Topics of the Course



Microbiology Topics

- ✓ Bacterial morphology and physiology
- ✓ Normal flora and opportunistic infections
- ✓ Bacterial infections and lab diagnosis
- ✓ Fungi and fungal diseases
- ✓ Viruses and viral infections



Infection Control Topics

- ✓ Disinfection and sterilization
- ✓ Antimicrobial chemotherapy
- ✓ Antimicrobial resistance
- ✓ Healthcare-associated infections (HAIs)
- ✓ Infection control policies



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Teaching Methods

You will learn through:



Lectures

- ✓ Lectures
- ✓ Group discussions
- ✓ Case studies
- ✓ Videos and diagrams



Learning is interactive, not only listening.

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Assessment & Grading System

Total Marks: 50



Final Written Exam: 40 marks



Week 15



Midterm Exam: 4 marks



Week 7



Quizzes: 4 marks



Weeks 5 & 11



Activities & Participation: 2 marks



Throughout the Semester



Continuous study and active participation are essential for success.



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Weekly Course Plan

Week	Topic
1	Introduction to Microbiology and Infection Control
2	Bacterial Morphology
3	Bacterial Physiology
4	Normal Flora & Opportunistic Infections
5	Disinfection and Sterilization + Quiz 
6	Antimicrobial Chemotherapy
7	Midterm Exam
8	Antimicrobial Resistance
9	Bacterial Infections & Laboratory Diagnosis
10	Fungi and Fungal Diseases
11	Viruses and Viral Infections + Quiz 
12	Healthcare-Associated Infections (HAIs)
13	Infection Control Policies in Healthcare Settings
14	Revision
15	Final Written Exam



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Student Responsibilities

What is expected from you?



Attend lectures regularly



Participate in discussions



Complete assignments



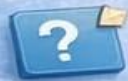
Follow infection control rules



Quiz



Use LMS and online resources



Ask questions when needed

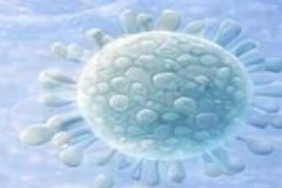


Nursing is teamwork — learning is also teamwork.





Learning Resources



Books

- ✓ Foundations in Microbiology
- ✓ Microbiology for Nurses



Websites

- ✓ **CDC** Infection Control
- ✓  WHO Infection Prevention
- ✓  Microbiology Online



Platforms

-  University LMS
-  Microsoft Teams
-  Google Classroom



Remember



Infection prevention saves lives.



Every nurse is responsible for patient safety.



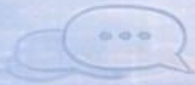
Microbiology is not just a subject — it is part of your daily clinical practice.



Be active. Be curious. Be professional.



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Thank You



Best Wishes for a Successful Semester



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Lecturer of Microbiology



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